Machine Learning-Based Diabetes Diagnosis and Hospital Decision-Support System Design Using the PEAS Framework

(Word count without figure/table/reference: 2999)

1 Abstract

This study developed a machine learning-based diabetes diagnosis workflow using a diabetes health dataset. Five classifiers were trained and compared: Random Forest, Gradient Boosting, Logistic Regression, Support Vector Machine, and K-Nearest Neighbors. Model performance was evaluated using confusion matrices, accuracy, precision, recall, specificity, F1-score, ROC-AUC, and PR-AUC. Gradient Boosting achieved the strongest overall performance, with the best performance and stability. Explainable machine learning methods, including coefficients, feature importance, SHAP, and permutation importance, consistently identified *FastingBloodSugar* and *HbA1c* as the dominant predictors. Finally, the PEAS framework was used to translate the classifier into a hospital diagnosis support system by defining performance goals, clinical environment, system outputs, and patient-data sensors.

2 Introduction

2.1 Diabetes and Machine Learning-Based Screening

Diabetes is a chronic metabolic disease associated with impaired glucose regulation and long-term complications such as cardiovascular disease, kidney damage, neuropathy, and visual impairment [1, 2]. Early identification of high-risk patients is clinically important because timely intervention can reduce disease progression and complications [2]. Since diabetes risk is influenced by clinical biomarkers, symptoms, lifestyle factors, medication use, and medical history, diagnosis can be treated as a multi-factorial binary classification task using machine learning methods [3].

2.2 PEAS Framework Introduction

Beyond prediction, this study also considered how the model could be implemented as a hospital diagnosis support system using the PEAS framework [4, 5]. Performance, Environment, Actuators, and Sensors were defined to connect the trained classifier with clinical workflow, data collection, risk reporting, follow-up recommendations, and clinician decision support.

3 Materials and Methods

3.1 Dataset

This study used a diabetes health dataset containing 1,879 patients [6]. Each record includes a wide range of patient-level variables covering demographics, lifestyle, medical history, clinical measurements, medication usage, symptoms, quality-of-life, environmental exposure and health behaviors. The prediction target was *Diagnosis*, a binary variable (0 for non-diabetes; 1 for diabetes). The dataset supports diabetes diagnosis as a multifactorial classification problem. The complete machine learning pipeline is provided in the accompanying Jupyter Notebook.

3.2 Data preprocessing and splitting

The raw dataset was first checked to ensure no abnormal values or columns. Multi-class categorical variables were converted using one-hot encoding so that category codes were not incorrectly treated as ordered ranks. Since the original dataset had already been released with pre-defined valid ranges for each variable, extreme values were not recognized as outliers.

An 80:20 train-test split was then applied, with stratification by *Diagnosis* to maintain the same diagnosis class proportions. Hyperparameter tuning was performed using stratified 5-fold cross-validation on the training set, providing more stable estimations of model performance.

Normalization was conducted for scale-sensitive models using equation (1), with the scaler fitted only on the training set and then applied to the testing set to avoid data leakage.

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

3.3 Machine Learning Models

Five classifiers were developed and compared: Random Forest (RF), Gradient Boosting (GB), Logistic Regression (LR), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). They were selected because they represent distinct modelling principles, allowing model performance comparison not only by accuracy, but also by interpretability, robustness and clinical suitability.

Random Forest was used as a bagging-based ensemble model. It trains multiple decision trees on bootstrap samples and combines their predictions by majority voting. This structure makes it able to capture non-linear relationships and be relatively robust to outliers. However, impurity-based feature importance may favor continuous variables with more possible split points, hence underestimates importance of categorical variables [7].

Gradient Boosting was included as a boosting-based tree ensemble. It builds weak learners sequentially, with each new tree correcting errors from previous trees. This gives strong predictive performance on structured datasets and learns complex non-linear patterns. However, it is sensitive to hyperparameter settings [8].

Logistic Regression served as an interpretable linear baseline. It models log-odds of diabetes as a linear combination of predictors and provides interpretable coefficients that indicate both direction and magnitude of association. However, it cannot easily capture non-linear effects or interactions, and its performance can be affected by feature scaling and correlated predictors [9].

SVM was used as a margin-based classifier. With a radial basis function kernel, it can model non-linear separating hyperplanes and perform well in high-dimensional feature spaces. However, it requires scaled features, its probability estimates are less direct, and its decision process is harder to interpret clinically [10].

KNN was included as a simple non-parametric instance-based comparator. It predicts each sample from the labels of nearby training samples, making few assumptions about the data distribution. However, it depends strongly on feature scaling, distance metric, and the selected number of neighbors [9].

3.4 Model Evaluation

Model evaluation was performed on the independent testing set using the best hyperparameters. Both threshold-dependent and threshold-independent metrics were reported (Table 1). Accuracy, precision, recall, specificity, and F1-score were calculated from the confusion matrix after selecting the threshold that maximized F1-score for each model. ROC-AUC and PR-AUC were also calculated from curves to assess model discrimination across thresholds.

3.5 Model Explanation

Explainable methods were applied to models to identify feature contributions. Four approaches were applied to specific models: coefficients, built-in feature importance, SHapley Additive exPlanations (SHAP) analysis and permutation importance. They were interpreted together as each method explains a different aspect of model behavior.

Logistic regression coefficients were used to examine linear effects. Since continuous variables were standardized before fitting the model, coefficient magnitudes were comparable across features. Positive and negative coefficients indicated whether a predictor pushed the model toward or away from diabetes. However, this interpretation remains limited to linear relationships between predictors and the log-odds of diagnosis. Additionally, L1 regularization was selected during hyperparameter tuning, thereby penalizing and shrinking some weaker predictors to zero, producing a relatively sparse linear model [9, 11].

Built-in feature importance was extracted from Random Forest and Gradient Boosting. This measure reflects how much each feature contributes to impurity reduction or informative splits across the ensemble. It is useful for ranking predictors in tree-based models, although it does not show the direction of feature effects and may favor continuous variables with more possible split points [12].

SHAP analysis was applied to Random Forest and Gradient Boosting to provide a richer explanation of tree-based predictions. SHAP values estimate how much each feature shifts an individual prediction relative to the model baseline. Therefore, SHAP can show both global importance and the direction of local feature effects, hence providing richer information. However, SHAP explains model behavior rather than causal relationships in the patient population [13].

Permutation importance was used as a model-agnostic explanation method. For each trained model, a single feature was randomly permuted, and the resulting decrease in ROC-AUC was measured. A larger decrease indicated stronger model dependence. Permutation importance can be directly linked to predictive performance and can be applied to various models using the same principle. However, it lacks directionality, and correlated features can complicate interpretation [14].

Metric	*Equation*
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	$\frac{TP}{TP + FP}$
Recall/Sensitivity/TPR	$\frac{TP}{TP + FN}$
Specificity/1-FPR	$\frac{TN}{TN + FP}$
F1-score	$2 \times \frac{Precision \times Recall}{Precision + Recall}$
ROC-AUC	$\int_0^1 TPR(FPR)d(FPR)$
PR-AUC	$\int_0^1 Precision(Recall)d(Recall)$

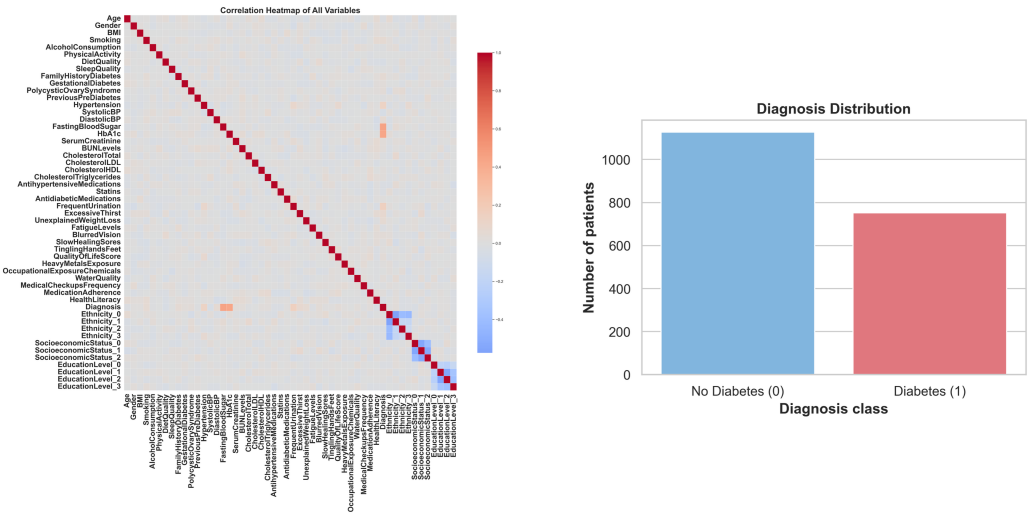
Table 1. *Equations for all calculated metrics in this study, TP, TN, FP, FN are extracted from confusion matrices for each model.*

4 Results

4.1 Exploratory Data Analysis (EDA)

The correlation heatmap showed that most predictors had weak pairwise correlations, suggesting no severe global multicollinearity (Figure 1A). Some negative correlations among one-hot encoded categories are not recognized as meaningful biological associations. The strongest correlations with *Diagnosis* were observed for *FastingBloodSugar* and *HbA1c*, with coefficients of 0.4520 and 0.4311 (Supplement Table S1). These two glucose-related biomarkers were therefore expected to provide strong predictive information. In contrast, most other variables showed weaker linear correlations with Diagnosis, suggesting that their contribution may depend on non-linear effects, feature interactions, or model-specific decision boundaries.

The target (*Diagnosis*) distribution showed a mild class imbalance. Among 1,879 patients, 1,127 were labelled as non-diabetic (59.98%) and 752 as diabetic (40.02%) (Figure 1B, Supplement Table S2). Although this imbalance was not severe, non-diabetic patients still formed the majority class. Therefore, stratified train-test splitting and stratified 5-fold cross-validation were used to preserve the class ratio during model evaluation and hyperparameter tuning. This also suggests importance of reporting precision, recall, specificity, F1-score, ROC-AUC and PR-AUC rather than relying on accuracy alone.



(A) **(B)**

Figure 1. *EDA of the processed dataset.* ***(A)*** *Correlation matrix for all variables. Most predictors had weak pairwise correlations, whereas FastingBloodSugar and HbA1c showed strongest correlations with Diagnosis.* ***(B)*** *Distribution of Diagnosis classes. Mildly imbalanced towards no diabetes (0).*

4.2 Model Training and Hyperparameter Tuning

Five classification models were trained and tuned using stratified 5-fold cross-validation with 50 randomized search iterations.

The hyperparameter search spaces were designed based on each algorithm's learning mechanism. For tree-based ensembles, parameters controlling model complexity, including tree depth, number of estimators, and minimum samples per split or leaf, were tuned to balance underfitting and overfitting. For Logistic Regression and SVM, regularization strength and class weight were explored because these models are sensitive to coefficient shrinkage and classification error. For KNN, the number of neighbors, distance weighting, and distance metric were tuned because these parameters directly determine local decision boundaries.

ROC-AUC was used as the optimization metric because it is independent of a single classification cut-off. Selected hyperparameters and corresponding ROC-AUC are summarized in Table 3.

Model	***Hyperparameter***	***Selected Value***	***ROC-AUC***
RF	*n_estimators*	*500*	*0.9549*
	max_depth	*10*	
	min_samples_split	*10*	
	min_samples_leaf	*4*	
	max_features	*sqrt*	
	bootstrap	*True*	
GB	*n_estimators*	*200*	*0.9601*
	learning_rate	*0.1*	

	max_depth	*2*	
	min_samples_split	*2*	
	min_samples_leaf	*2*	
	subsample	*1.0*	
	max_features	*sqrt*	
LR	*C*	*0.1*	*0.9029*
	penalty	*l1*	
	solver	*liblinear*	
	class_weight	*balanced*	
	fit_intercept	*True*	
SVM	*C*	*10*	*0.8978*
	kernel	*rbf*	
	gamma	*0.01*	
	class_weight	*balanced*	
KNN	*n_neighbors*	*31*	*0.8615*
	weights	*uniform*	
	metric	*minkowski*	
	algorithm	*ball_tree*	
	p	*1*	

Table 3. *Selected hyperparameters for each model after iterative tuning, together with ROC-AUC they achieved. Using optimized hyperparameters, gradient boosting achieved highest ROC-AUC at 0.9601*

4.3 Model Evaluation

The tuned models were evaluated on the independent testing set. For each model, predicted probabilities were converted into class labels using the threshold that maximized F1-score, and the resulting confusion matrices are summarized in Table 4. Calculated metrics were summarized in Table 5.

In diabetes screening, false negatives are clinically more important than false positives. These patients may miss timely follow-up, lifestyle intervention, or treatment. Therefore, interpretation should pay particular attention to false-negative counts. Also, recall was treated as a key metric. Precision and specificity were also considered, because a model that labels too many non-diabetic patients as diabetic would create unnecessary follow-up tests and clinical workload.

The metric table showed that Gradient Boosting achieved the strongest overall performance. It obtained the best values across main evaluation metrics. Its high recall indicated a low false-negative rate, which is particularly important in diabetes screening, while its high accuracy showed strong overall prediction reliability. The radar chart (Figure 2C) further supported this result: Gradient Boosting covered the widest and most balanced area across the evaluation metrics, exceeding overall profile of other models.

The other models still provided useful comparisons. Random Forest performed very

close to Gradient Boosting and remained the strongest alternative. SVM and Logistic Regression showed moderate but stable performance. KNN showed a different profile: although its precision and specificity were relatively low, its recall stands out, which is suitable for the screening task, hence also considered as a usable model. The ROC and precision-recall curves (Figure 2A, B) showed reasonable curve shapes and AUC values for all five models, suggesting that each model learned meaningful classification patterns.

Model stability was then assessed by comparing the average ROC-AUC from stratified 5-fold cross-validation with the test ROC-AUC (Figure 2D, Supplement Table S3). The comparison showed no major decrease from cross-validation to testing for any model, suggesting that performance was largely maintained under random data splitting. Gradient Boosting again showed the strongest overall stability.

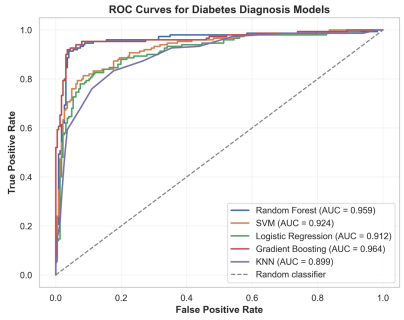
Overall, Gradient Boosting was recommended as the preferred model due to its superior performance and stability. However, other models also retained practical value.

Model	***TN***	***FP***	***FN***	***TP***
RF	*217*	*9*	*13*	*137*
GB	*218*	*8*	*12*	*138*
LR	*209*	*17*	*33*	*117*
SVM	*207*	*19*	*28*	*122*
KNN	*186*	*40*	*25*	*125*

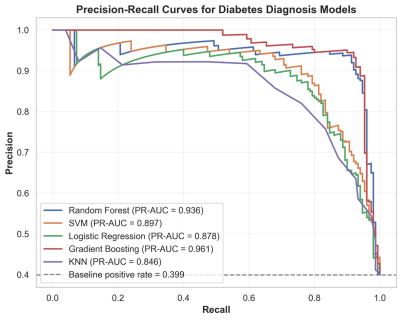
Table 4. *Summarized confusion matrices for models. Gradient Boosting showed smallest FN and FP.*

Model	***Threshold***	***Accuracy***	***Precision***	***Recall***	***Specificity***	***F1-score***	***ROC-AUC***	***PR-AUC***
RF	*0.4169*	*0.9415*	*0.9384*	*0.9133*	*0.9602*	*0.9257*	*0.9589*	*0.9361*
GB	*0.3749*	*0.9468*	*0.9452*	*0.9200*	*0.9646*	*0.9324*	*0.9636*	*0.9611*
LR	*0.6167*	*0.8670*	*0.8731*	*0.7800*	*0.9248*	*0.8239*	*0.9116*	*0.8776*
SVM	*0.5216*	*0.8750*	*0.8652*	*0.8133*	*0.9159*	*0.8385*	*0.9245*	*0.8968*
KNN	*0.4194*	*0.8271*	*0.7576*	*0.8333*	*0.8230*	*0.7937*	*0.8988*	*0.8457*

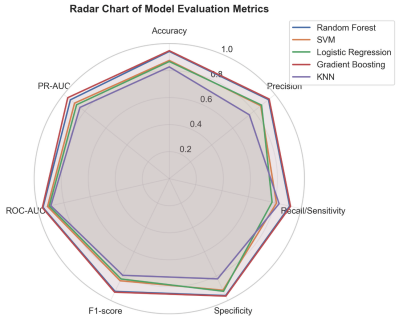
Table 5. *Summarized metrics for models. Gradient Boosting achieved overall high performance.*



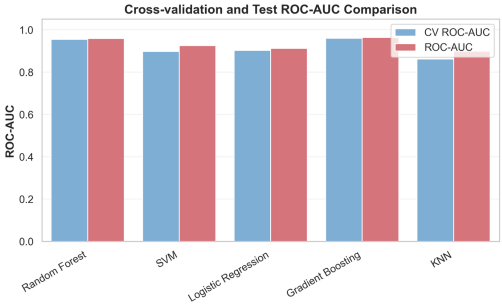
(A)



(B)



(C)



(D)

Figure 2. *Performance comparison among models.* ***(A)*** *ROC curves for models, evaluating how well a model separates diabetic and non-diabetic patients across different classification thresholds. High AUC ensures model robustness.* ***(B)*** *PC curves for models evaluate how reliably they detect diabetic patients by showing the trade-off between precision and recall. High AUC ensures model robustness.* ***(C)*** *Radar plot compares models' performance. High coverage rate of Gradient Boosting showed its overall high performance.* ***(D)*** *Bar plot compares average ROC-AUC during 5-fold CV and highest ROC-AUC. Slight decrease in values showed stability of all models.*

4.4 Model Explanation

4.4.1 Logistic Regression Coefficients

Logistic Regression coefficients were examined to identify predictors with strong linear effects on diabetes. The strongest positive coefficients were observed for *FastingBloodSugar*, *HbA1c*, *FrequentUrination,* and *Hypertension*, with coefficients larger than 1 (Figure 3A, Supplement Table S4). Some features, such as *UnexplainedWeightLoss* at category 0 (Low) and *SystolicBP,* showed negative coefficients, reducing the model's predicted probability of diabetes (Figure 3B, Supplement Table S4).

4.4.2 Feature importance

Built-in feature importance was extracted from Random Forest and Gradient Boosting to identify predictors most frequently used in informative tree splits (Figure 3C, D, Supplement Table S5, S6). Both tree-based models identified *FastingBloodSugar* and *HbA1c* as the dominant predictors.

4.4.3 SHAP analysis

SHAP analysis was also applied to the two tree-based models. SHAP values measure how much

each feature shifts an individual prediction away from the model baseline. Positive SHAP values pushed the prediction toward diabetes, while negative values shifted it toward the non-diabetic class. The SHAP summary plots showed consistent attribution patterns across Random Forest and Gradient Boosting (Figure 3E, F, Supplement Table S7, S8). In both models, *FastingBloodSugar* and *HbA1c* were ranked as the two most influential predictors, with the widest SHAP value distributions. This indicates that these two biomarkers contributed the largest changes to individual diabetes predictions in a monotonic pattern. *FrequentUrination* and *Hypertension* (binary variables) also showed clear separated clusters, and bring positive contributions when present, meaning that patients with these symptoms or conditions were more likely to be classified as diabetic by the tree-based models.

Considering Gradient Boosting was selected as our most preferred model, its SHAP dependence plots were examined in more detail (Figure 3 G, H). *FastingBloodSugar* and *HbA1c* was extracted for plotting, and showed clear threshold-like patterns. At lower values, approximately below 130 mg/dL for *FastingBloodSugar* and below 6.5% for *HbA1c*, SHAP values were mostly negative. Above these ranges makes SHAP values shift toward positive values.

4.4.4 Permutation Importance

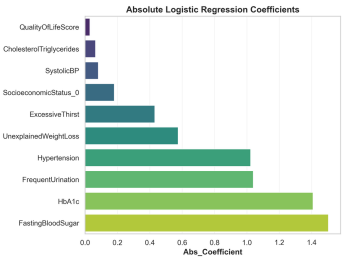
Permutation importance was used as a model-agnostic explanation method (Figure 3I, Supplement Table S9). The permutation importance heatmap showed that *HbA1c* and *FastingBloodSugar* still remained important across all five models by causing the largest decreases in ROC-AUC. *FrequentUrination* and *Hypertension* also appeared as secondary contributors with smaller importance values.

4.4.5 Combined Interpretation

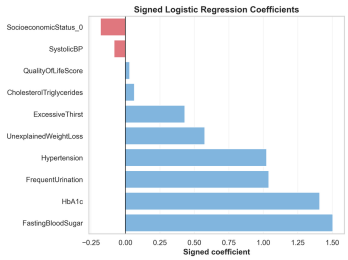
The explanation methods produced a consistent pattern: *FastingBloodSugar* and *HbA1c* were recognized as the dominant predictors, suggesting that the models relied mainly on direct glycaemic indicators rather than weaker contextual variables. This is expected because fasting blood sugar and HbA1c are core measurements of short-term and longer-term glucose status [1].

Secondary predictors, including *FrequentUrination* and *Hypertension,* also added supporting information. These features are still clinically relevant because they describe symptoms or comorbid conditions often associated with diabetes [15].

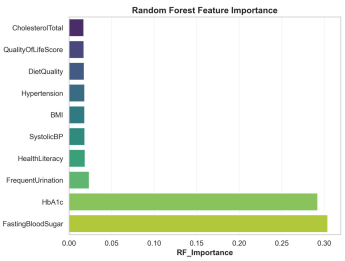
Overall, the agreement across explanations strengthened the reliability of the interpretation. However, these results should be understood as predictive evidence rather than causal proof, since the models explain classification behavior within this dataset rather than disease mechanisms.


Absolute Logistic Regression Coefficients
QualityOfLifeScore
CholesterolTriglycerides
SystolicBP
SocioeconomicStatus_0
ExcessiveThirst
UnexplainedWeightLoss
Hypertension
FrequentUrination
HbA1c
FastingBloodSugar
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4
Abs_Coefficient

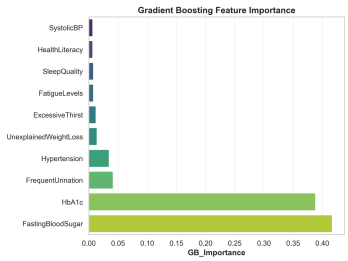
(A)


Signed Logistic Regression Coefficients
SocioeconomicStatus_0
SystolicBP
QualityOfLifeScore
CholesterolTriglycerides
ExcessiveThirst
UnexplainedWeightLoss
Hypertension
FrequentUrination
HbA1c
FastingBloodSugar
−0.25 0.00 0.25 0.50 0.75 1.00 1.25 1.50
Signed coefficient

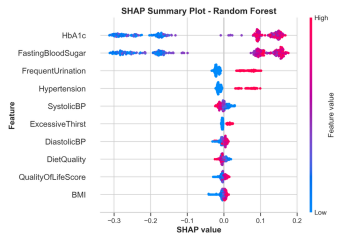
(B)


Random Forest Feature Importance
CholesterolTotal
QualityOfLifeScore
DietQuality
Hypertension
BMI
SystolicBP
HealthLiteracy
FrequentUrination
HbA1c
FastingBloodSugar
0.00 0.05 0.10 0.15 0.20 0.25 0.30
RF_Importance

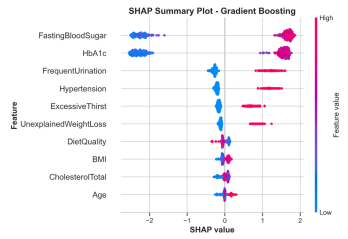
(C)


Gradient Boosting Feature Importance
SystolicBP
HealthLiteracy
SleepQuality
FatigueLevels
ExcessiveThirst
UnexplainedWeightLoss
Hypertension
FrequentUrination
HbA1c
FastingBloodSugar
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40
GB_Importance

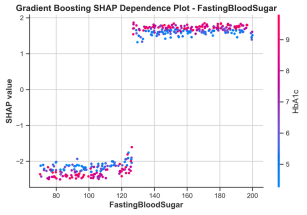
(D)


SHAP Summary Plot - Random Forest
HbA1c
FastingBloodSugar
FrequentUrination
Hypertension
SystolicBP
ExcessiveThirst
DiastolicBP
DietQuality
QualityOfLifeScore
BMI
Feature
−0.3 −0.2 −0.1 0.0 0.1 0.2
SHAP value
High
Feature value
Low

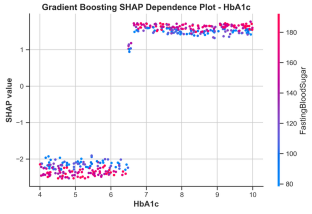
(E)


SHAP Summary Plot - Gradient Boosting
FastingBloodSugar
HbA1c
FrequentUrination
Hypertension
ExcessiveThirst
UnexplainedWeightLoss
DietQuality
BMI
CholesterolTotal
Age
Feature
−2 −1 0 1 2
SHAP value
High
Feature value
Low

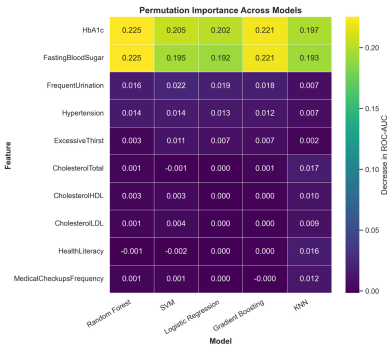
(F)


Gradient Boosting SHAP Dependence Plot - FastingBloodSugar
SHAP value
80 100 120 140 160 180 200
FastingBloodSugar
HbA1c

(G)


Gradient Boosting SHAP Dependence Plot - HbA1c
SHAP value
4 5 6 7 8 9 10
HbA1c
FastingBloodSugar

(H)


Permutation Importance Across Models
HbA1c 0.225 0.205 0.202 0.221 0.197
FastingBloodSugar 0.225 0.195 0.192 0.221 0.193
FrequentUrination 0.016 0.022 0.019 0.018 0.007
Hypertension 0.014 0.014 0.013 0.012 0.007
ExcessiveThirst 0.003 0.011 0.007 0.007 0.002
CholesterolTotal 0.001 -0.001 0.000 0.001 0.017
CholesterolHDL 0.003 0.003 0.000 0.000 0.010
CholesterolLDL 0.001 0.004 0.000 0.000 0.009
HealthLiteracy -0.001 -0.002 0.000 0.000 0.016
MedicalCheckupsFrequency 0.001 0.001 0.000 -0.000 0.012
Feature
Random Forest
SVM
Logistic Regression
Gradient Boosting
KNN
Model
0.20
0.15
0.10
0.05
0.00
Decrease in ROC-AUC

(I)

Figure 3. *Results of model explanation methods. Patterns are relatively consistent across methods and models.* ***(A)-(B)*** *Absolute Logistic Regression coefficients, color represents the magnitude of coefficients & Signed Logistic Regression coefficients, red bars are negative coefficients, while blue bars are positive.* ***(C)-(D)*** *Build-in feature importance for Random Forest and Gradient Boosting.* ***(E)-(F)*** *SHAP summary plots for Random Forest and Gradient Boosting positive SHAP values lead to prediction of diabetes, while negative values lead to non-diabetes.* ***(G)-(H)*** *SHAP dependence plot for Gradient Boosting's top 2 importance variables: FastingBloodSugar with threshold of around 130 mg/dL and HbA1c with threshold of around 6.5%.* ***(I)*** *Permutation importance for all five models, summarized into a heatmap. Color represents the magnitude of importance.*

5. Discussion

5.1 PEAS Framework for Diabetes Diagnosis System

The proposed classifier was further translated into a hospital diabetes diagnosis support system using the PEAS framework. This connects the machine learning results with a practical clinical workflow, defining how the system should be evaluated, where it operates, what outputs it produces, and what patient data it requires.

5.2 Performance

Performance should cover both diagnostic quality and clinical usability. At the diagnostic level, the system should be assessed using metrics such as accuracy, precision, recall, specificity, F1-score, ROC-AUC, and PR-AUC. Beyond prediction quality, performance should also include response speed, integration into routine consultations or health checks, clinician and patient satisfaction, follow-up rates, earlier identification of high-risk patients, and post-intervention outcomes. For deployment, computational cost, storage demand, and system efficiency should also be monitored to ensure that the tool remains practical in hospital settings.

5.3 Environment

The proposed system operates in a complex clinical environment. Patient status is only partially observable because not all symptoms, behaviors, or historical records are complete. The environment is also stochastic, since patients and clinicians may not always follow system recommendations exactly. It is sequential because past medical history, current measurements, and future follow-up all affect decision-making. It is semi-dynamic because patient condition may not change that fast, while system response time remains part of performance. The system is also continuous for numeric percepts, and multi-agent, involving patients, clinicians, nurses, and administrative staff.

In practice, the system would operate in hospital outpatient clinics, primary-care screening services, or routine health-check settings. It should integrate with electronic health record systems and support users who need rapid risk assessment during patient evaluation. The patient population is clinically heterogeneous, with variation in age, lifestyle, medical history, medication use, symptoms, and laboratory results. Therefore, the system must process multi-source clinical data rather than rely on a single biomarker. Its role should remain decision support: it can prioritize risk, highlight key predictors, and suggest follow-up, but the final diagnosis should stay under clinician judgment, especially when model output conflicts with laboratory evidence or patient history.

5.4 Actuators

The system's actuators are the outputs delivered to clinicians, patients, and administrative staff. For clinicians, the system should generate a

structured risk report containing predicted diabetes status, risk probability, risk category, and key contributing factors. For high-risk patients, automated alerts could recommend further laboratory confirmation, specialist referral, or earlier follow-up. The explanation module should highlight clinically interpretable drivers, such as elevated HbA1c or fasting blood sugar, so that clinicians can judge whether the prediction is consistent with the patient's condition. For patients, the system should provide understandable feedback and personalized education, including advice on diet quality, physical activity, medication adherence, and regular check-ups. For administrative or technical staff, the system could support follow-up scheduling, threshold monitoring, model performance tracking, and database management.

5.5 Sensors

Sensors define the data collected by the system. In this project, the model inputs correspond to realistic clinical data sources. Patient questionnaires can collect demographics, lifestyle factors, symptoms, health behaviors, and quality-of-life scores. Clinical measurement devices can provide BMI and blood pressure, while laboratory systems can supply fasting blood sugar, HbA1c, serum creatinine, BUN, and cholesterol profiles. Electronic health records can contribute family history, previous pre-diabetes, hypertension, medication use and other comorbidities. Environmental and behavioral records, including heavy metal exposure, occupational chemical exposure, water quality, medical check-up frequency, medication adherence, and health literacy, provide additional context. Since *FastingBloodSugar* and *HbA1c* were consistently identified as the dominant predictors, reliable laboratory data collection should be treated as essential for system performance.

6 Conclusion

This project developed and compared five machine learning models for binary diabetes diagnosis using a multi-dimensional patient health dataset. All models exceeded the required 80% accuracy target. Gradient Boosting showed the highest performance and stability. It was therefore recommended as the best model. However, the other models also retained practical value, and multiple models could be used together as reference models or combined through a voting strategy to improve robustness.

Model explanation results were consistent across cmethods. *FastingBloodSugar* and *HbA1c* were the dominant predictors. This pattern supports the clinical relevance of the trained models, since the strongest predictors align with glucose regulation and diabetes-related symptoms.

The PEAS framework further translated the classifier into a hospital decision-support design, specifying performance goals, clinical environment, system outputs, and patient-data sources.

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